Algorithms

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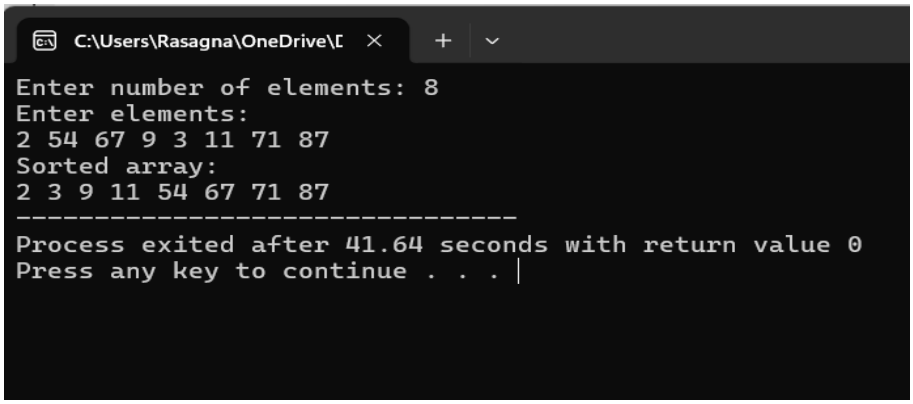
LAB – 1 .

Bubble Sort C Code:

```
#include <stdio.h>
int main() {
   int n, i, j, temp;
   printf("Enter number of elements: ");
   scanf("%d", &n);
   int arr[n];
   printf("Enter elements:\n");
   for(i = 0; i < n; i++) {
      scanf("%d", &arr[i]);
   }
   for(i = 0; i < n - 1; i++) {
      for(j = 0; j < n - i - 1; j++) {
         if(arr[j] > arr[j + 1]) {
            temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
         }
      }
```

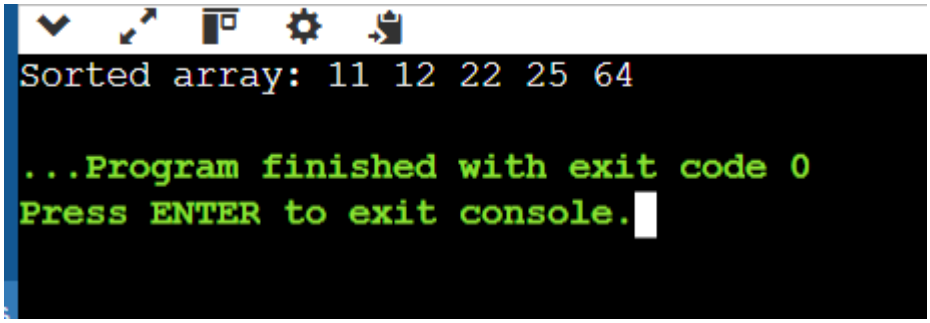
```
    }
    printf("Sorted array:\n");
    for(i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    return 0;
}
```



Selection C Code:

```
#include <stdio.h>
int main() {
    int arr[] = {64, 25, 12, 22, 11};
    int n = 5;
    int i, j, min_index, temp;
    for (i = 0; i < n - 1; i++) {
        min_index = i;
        for (j = i + 1; j < n; j++) {
            if (arr[j] < arr[min_index]) {
                min_index = j;
            }
```

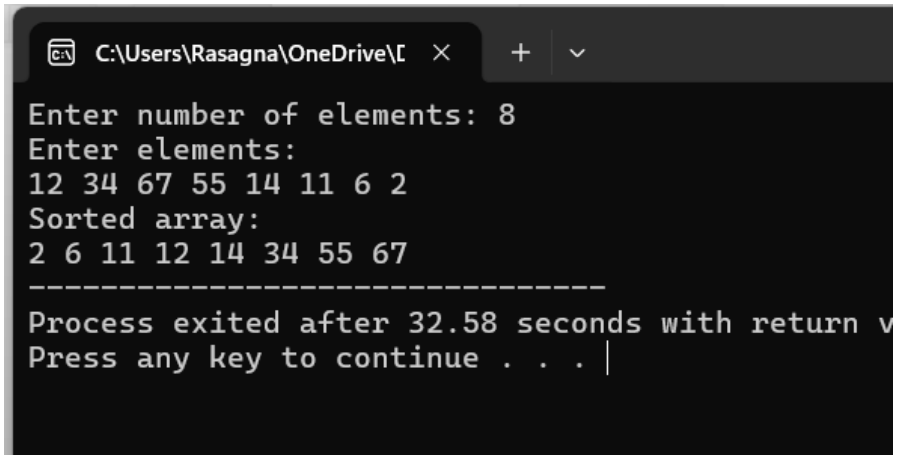
```
        }
        temp = arr[i];
        arr[i] = arr[min_index];
        arr[min_index] = temp;
    }
    printf("Sorted array: ");
    for (i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    return 0;
}
```



Insertion sort C Code:

```
#include <stdio.h>
int main() {
    int n, i, j, key;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
```

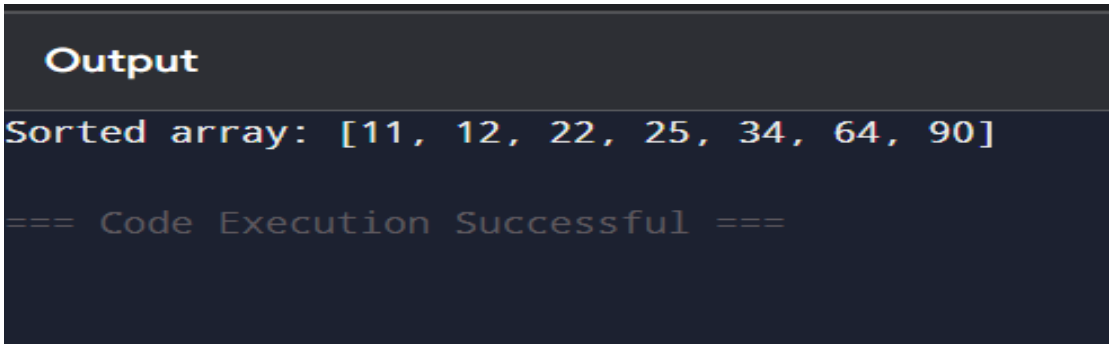
```
    }
    for(i = 1; i < n; i++) {
        key = arr[i];
        j = i - 1;
        while(j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
    printf("Sorted array:\n");
    for(i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    return 0;
}
```



Bubble Sort Python Code:

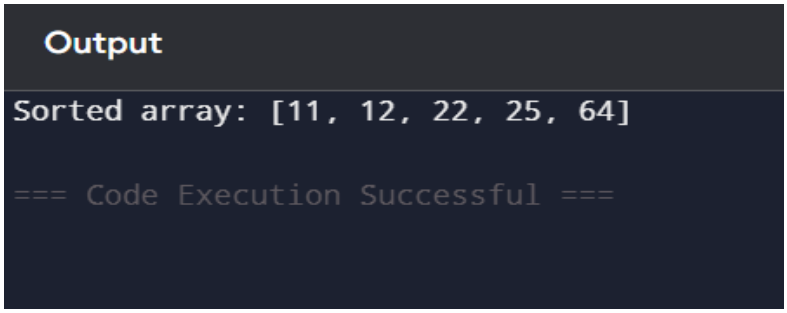
```
arr = [64, 34, 25, 12, 22, 11, 90]
n = len(arr)
```

```
for i in range(n - 1):
    for j in range(n - i - 1):
        if arr[j] > arr[j + 1]:
            arr[j], arr[j + 1] = arr[j + 1], arr[j]
print("Sorted array:", arr)
```



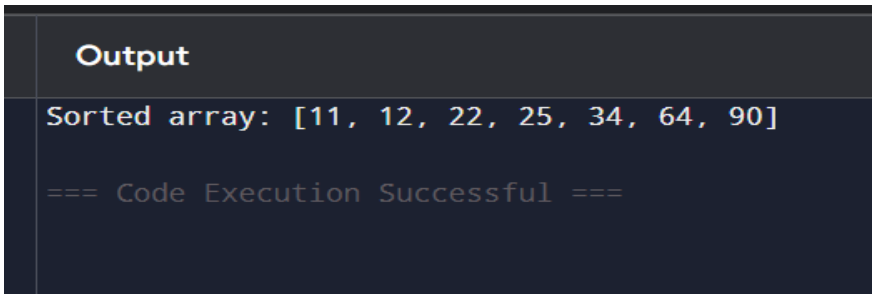
Selection Python code

```
def selection_sort(arr):
    n = len(arr)
    for i in range(n - 1):
        min_index = i
        for j in range(i + 1, n):
            if arr[j] < arr[min_index]:
                min_index = j
        arr[i], arr[min_index] = arr[min_index], arr[i]
arr = [64, 25, 12, 22, 11]
selection_sort(arr)
print("Sorted array:", arr)
```



Insertion sort Python code:

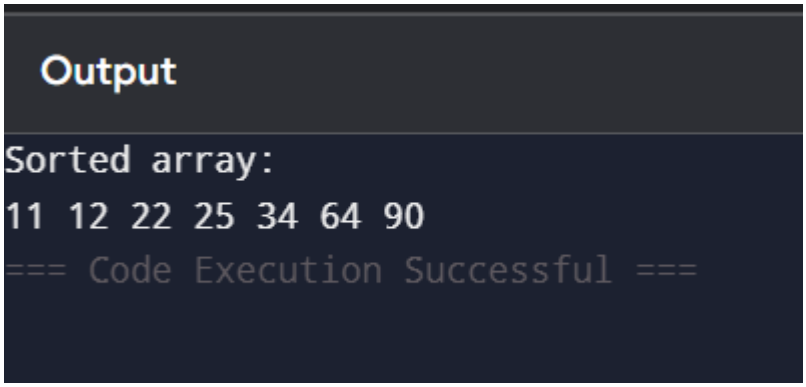
```
arr = [64, 34, 25, 12, 22, 11, 90]


for i in range(1, len(arr)):
    key = arr[i]
    j = i - 1
    while j >= 0 and arr[j] > key:
        arr[j + 1] = arr[j]
        j -= 1
    arr[j + 1] = key
print("Sorted array:", arr)
```



Bubble Sort Java Code:

```
public class BubbleSort {
    public static void main(String[] args) {
        int[] arr = {64, 34, 25, 12, 22, 11, 90};
        for (int i = 0; i < arr.length - 1; i++) {
```

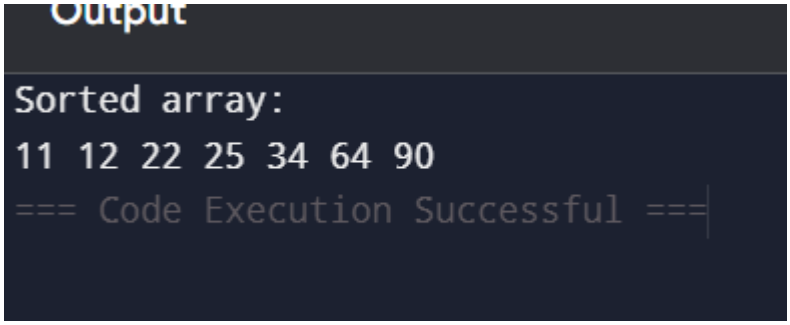
```
        for (int j = 0; j < arr.length - i - 1; j++) {


            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    System.out.println("Sorted array:");
    for (int i = 0; i < arr.length; i++) {
        System.out.print(arr[i] + " ");
    }
  }
}
```



Insertion Sort Java Code:

```
public class InsertionSort {
    public static void main(String[] args) {
        int[] arr = {64, 34, 25, 12, 22, 11, 90};
        for (int i = 1; i < arr.length; i++) {
```

```
        int key = arr[i];
        int j = i - 1;
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
    System.out.println("Sorted array:");
    for (int i = 0; i < arr.length; i++) {
        System.out.print(arr[i] + " ");
    }
  }
}
```



Selection sort Java Code:

```
public class SelectionSort {
  public static void main(String[] args) {
    int[] arr = {64, 25, 12, 22, 11};
    for (int i = 0; i < arr.length - 1; i++) {
      int minIndex = i;
      for (int j = i + 1; j < arr.length; j++) {
        if (arr[j] < arr[minIndex]) {
```

```
                minIndex = j;
            }
        }
        int temp = arr[i];
        arr[i] = arr[minIndex];
        arr[minIndex] = temp;
    }
    System.out.print("Sorted array: ");
    for (int i = 0; i < arr.length; i++) {
        System.out.print(arr[i] + " ");
    }
  }
}
```